

PCB Surface Finishes

HASL • Lead-Free HASL • ENIG • OSP • Immersion Silver • Hard Gold

The thin layer that decides solder joint quality, shelf life, and cost

Audience: Intermediate Engineering Students

Selection • Cost • Pad flatness • Shelf life • Process compatibility

Why Surface Finish Matters

It's between your copper and the rest of the world

Solderability

Determines how well solder wets the pad.

Oxidation protection

Bare Cu oxidises in days
Finish extends shelf life.

Pad flatness

Critical for fine-pitch BGA
(≤ 0.5 mm pitch).

Cost

HASL cheapest, ENIG 3×
Hard gold 5-10×.

Reliability

ENIG can suffer 'black pad'
OSP requires care.

Compliance

HASL has lead — banned in EU
Lead-free finishes required.

Finish Comparison Table

All common finishes at a glance

Finish	Cost	Flatness	Shelf life	Best for
HASL (Sn-Pb)	₹ (1.0×)	Poor	12+ months	Through-hole, large SMD
Lead-free HASL (SAC)	₹ (1.1×)	Poor	12+ months	RoHS retrofit of legacy
ENIG	₹₹₹ (3×)	Excellent	12+ months	Fine-pitch BGA, default modern
OSP	₹ (1.0×)	Excellent	6 months	Short-cycle products
Immersion Silver	₹₹ (1.5×)	Excellent	6 months	RF, high-frequency
Immersion Tin	₹₹ (1.5×)	Excellent	6 months	Some Asian markets
Hard Gold (electrolytic)	₹₹₹₹ (5×)	Excellent (thick)	Years	Edge connectors, switches
ENEPIG	₹₹₹₹ (4×)	Excellent	12+ months	Aluminium wire bond + solder

HASL — Hot Air Solder Levelling

The classic, cheapest finish

- ▶ Bare PCB dipped in molten Sn-Pb (or lead-free SAC305)
- ▶ Hot air knife blows excess solder off → leaves coating on pads
- ▶ Coating thickness: 1-25 μm — uneven, somewhat bumpy
- ▶ Solderability: excellent (already covered in solder!)
- ▶ Shelf life: 12+ months
- ▶ Cost: cheapest of all finishes

Pros

Cheapest possible
Great for through-hole
Long shelf life
Forgiving of poor solder profiles

Cons

UNEVEN — bad for BGA + fine-pitch
Process uses molten solder (hot, costly)
Lead variant banned in EU (RoHS)
Lead-free HASL has issues with copper dissolution

Lead-Free HASL (SAC305)

RoHS-compliant version of classic HASL

- ▶ Same process as HASL — but uses SAC305 alloy (96.5% Sn / 3% Ag / 0.5% Cu)
- ▶ Higher process temperature (~250 °C vs ~210 °C for Sn-Pb)
- ▶ Stresses the PCB more — needs High-Tg FR-4
- ▶ More copper dissolution into the bath — pad damage if mishandled
- ▶ RoHS-compliant — sell in EU / global markets
- ▶ Cost: ~10% higher than Sn-Pb HASL
- ▶ Used as a drop-in replacement for legacy Sn-Pb HASL boards
- ▶ Not as flat as ENIG or OSP — still avoid for fine-pitch BGA

ENIG — Electroless Nickel Immersion Gold

Today's industry default for non-trivial boards

- ▶ Two-layer plating: 3-6 μm electroless Ni, then 50-100 nm immersion Au
- ▶ Gold protects nickel from oxidation
- ▶ During reflow, gold dissolves into solder; nickel forms the joint
- ▶ FLAT surface — perfect for fine-pitch BGA, QFN, leadless packages
- ▶ Solderability: excellent and consistent
- ▶ Shelf life: 12+ months

Famous failure mode: black pad

If nickel layer over-oxidises during the immersion gold step, joints become brittle, fail under stress, look DULL/BLACK. Modern processes have largely eliminated this — but pick reputable fabs.

Cost & use

~3× HASL cost. Modern default for any board with QFN / BGA / fine pitch. Nickel barrier also good for repeated reflow cycles.

OSP — Organic Solderability Preservative

Cheap, flat, short shelf life

- ▶ Thin organic coating ($\sim 0.2 \mu\text{m}$) bonded to copper — colourless or yellow
- ▶ Burns off during reflow → exposes fresh Cu just in time for solder wetting
- ▶ Cost: similar to HASL — among the cheapest
- ▶ FLAT surface — good for fine pitch
- ▶ Drawback: SHORT shelf life (6 months max), and limited to 2-3 reflow cycles
- ▶ Easily damaged by handling — needs careful storage in moisture-sealed bags
- ▶ Used in: short-cycle consumer products (phones, TVs) where assembly happens fast after fab
- ▶ If your boards sit on a shelf > 6 months before assembly → don't use OSP

Immersion Silver

Flat, low-cost, excellent for RF

- ▶ Pure silver layer (0.1-0.4 μm) immersion-plated onto copper
- ▶ FLAT surface — great for fine pitch + RF (low loss)
- ▶ Lower contact resistance than gold — preferred for RF connectors
- ▶ Cost: 1.5× HASL
- ▶ Shelf life: 6 months sealed; tarnishes in sulfur-containing environments
- ▶ Not common in India — more popular in NA / Europe

RF advantage

Silver has the lowest electrical resistance of any metal. At high frequencies, skin effect makes surface conductivity critical. Immersion silver is the choice for RF stripline circuits.

Tarnish caution

Silver tarnishes in H_2S environments (industrial pollution, sulfurous rubber). Storage matters. Anti-tarnish bags + low humidity.

Hard Gold (Electrolytic)

Thick, wear-resistant — for contact pads

- ▶ Electrolytic plating: thick gold (10-50 μin = 0.25-1.25 μm) on top of nickel barrier
- ▶ Hardened with Ni or Co alloy — resists wear from repeated contact
- ▶ Used on: edge connectors (PCIe cards, memory DIMMs), keyboard switches, push-buttons
- ▶ NOT for soldering — gold dissolves into solder, embrittles the joint
- ▶ Specify SELECTIVELY — only where contact wear matters, NOT on solder pads
- ▶ Cost: 5-10× HASL for the gold-only areas (must mask the rest)
- ▶ Gold plating thickness measured in μin (micro-inches) — typical 15-30 μin

ENEPIG — Ni / Pd / Au

Premium finish for mixed assembly

- ▶ Three layers: 3-6 µm electroless Ni, 0.05-0.3 µm electroless Pd, 50-100 nm Au
- ▶ Palladium prevents the 'black pad' issue with ENIG
- ▶ FLAT — like ENIG
- ▶ Compatible with: SMT solder, aluminium wire bond, gold wire bond
- ▶ Used in: high-rel products, mixed wire-bond + SMT assemblies
- ▶ Cost: ~4× HASL — premium-tier

Why three layers

Pd (palladium) barrier prevents nickel from oxidising, AND enables Al wire bonding (which ENIG can't reliably do). Aerospace & defense favourite.

When to spec

If you have BOTH SMD parts AND aluminium-wire-bond ICs (rare today), or high-reliability requirement and budget allows.

Process Compatibility

What finishes work with what assembly

Finish	SMT reflow	Wave soldering	Press-fit	Wire bond	Cycles
HASL	✓	✓	✓	✗	Many
Lead-free HASL	✓	✓	✓	✗	Many
ENIG	✓	✓	✓	Au (gold) wire	3-5
OSP	✓	Limited	✗	✗	2-3
Immersion Ag	✓	✓	✓	✗	3-5
Hard Gold	Avoid	Avoid	Edge only	Au or Al wire	Many
ENEPIG	✓	✓	✓	Au + Al wire	Many

Selection Workflow

Pick the right finish in 3 questions

- ▶ 1. Smallest part pitch on the board?
 - ▶ • > 1.27 mm pitch (large SOIC, THT) → HASL is fine
 - ▶ • 0.5-1.27 mm pitch (TQFP, QFN) → ENIG or OSP
 - ▶ • < 0.5 mm pitch (BGA, fine QFN) → ENIG (flatness critical)
- ▶ 2. Shelf life before assembly?
 - ▶ • < 6 months → OSP cheapest
 - ▶ • > 6 months → ENIG or HASL
- ▶ 3. Compliance / region?
 - ▶ • EU / RoHS strict → no leaded HASL
 - ▶ • RF / high-frequency → immersion silver
 - ▶ • Edge connectors → hard gold (selective)

Common Surface Finish Mistakes

Lessons from production

- ▶ HASL on BGA: uneven pad height → poor ball seating → opens
- ▶ OSP + 6+ month shelf wait → boards arrive poorly solderable
- ▶ Hard gold on solder pads: brittle joints, fail under thermal stress
- ▶ ENIG + cheap fab → black pad failures, intermittent shorts
- ▶ Specifying HASL for EU shipment → RoHS rejection
- ▶ OSP + multiple reflow cycles → film exhausted, poor wetting
- ▶ Not specifying gold thickness for edge connectors → wear-out fast
- ▶ Mixing finishes (ENIG main + hard gold edge) — fab can do but adds cost

Real Cost Impact

On a typical 50×50 mm 4-layer board, 100 pcs

Finish	Cost per board (₹)	Multiplier vs HASL
HASL (Sn-Pb)	100	1.0×
Lead-free HASL (SAC305)	110	1.1×
OSP	100	1.0×
Immersion Silver	150	1.5×
ENIG (3-6 μin Au)	300	3.0×
ENIG (50 μin Au)	450	4.5×
Hard Gold (edge only)	200 + ₹15/contact	2.0× + adders
ENEPIG	400	4.0×

Key Takeaways

Key takeaways from this lecture

ENIG = modern default

Flat, reliable, BGA-friendly.

HASL = legacy

Cheapest, but bumpy.

OSP = short cycle

Cheap + flat but 6-month limit.

Silver = RF specialty

Lowest loss for stripline.

Hard Gold = wear pads only

Brittle in solder joints.

Black pad = real risk

Pick reputable ENIG fab.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-4552 / 4553 — ENIG / Immersion Silver specifications
- ▶ IPC-4561 — OSP specification
- ▶ JLCPCB / PCBWay surface finish guide
- ▶ DuPont app notes on hard gold edge plating
- ▶ Atotech / Coventya surface finish chemistry
- ▶ Phil's Lab — surface finish comparison videos
- ▶ Black pad failure analysis case studies (online)
- ▶ Local fab capabilities — confirm before specifying

Thank you • Build something this week.